

Online Internship on Open-Source Processor Architecture (IBM OpenPOWER) and SoC Design

Assignment No. 02: Combinational Logic Design

| Instructions to Students

Please upload the complete compiled code, simulation screenshots, and documentation compiled as a single PDF file.

| Assignment Questions

Q1. Design and Simulate 8:3 Encoder. Simulate all possible input combinations and verify the output.

Q2. Design and simulate 4-bit magnitude comparator that generates outputs indicating:

- $A > B$
- $A = B$
- $A < B$

Q3. Design and simulate 3-bit Binary-to-Gray code converter and verify its operation through simulation.